BEAMLINE

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DESIGNED IN BEAMLINE



Alloptical Half Adder

Overview

All-optical half adder in a single chi(2) crystal (published PPLN scheme): A (1064 nm) and B (1550 nm) lasers are overlapped and sent through one quasi-phase-matched PPLN waveguide. CARRY = the 631 nm sum-frequency light (A AND B). SUM = the residual infrared: with matched photon fluxes and strong conversion, both inputs are nearly fully depleted when both are on, so the leftover IR is bright for exactly one input and dim for both (A XOR B). Default state is 1 + 1 = 10.

Photon fluxes matched (P_A/P_B = 1550/1064, 218.5 / 150.0 mW) so both inputs deplete together. eta = 50 /W (5000 %/W) is an optimistic long PPLN ridge waveguide. Default: A = B = 1 -> SUM 0, CARRY 1 (1 + 1 = 10). Logic threshold 40 mW.

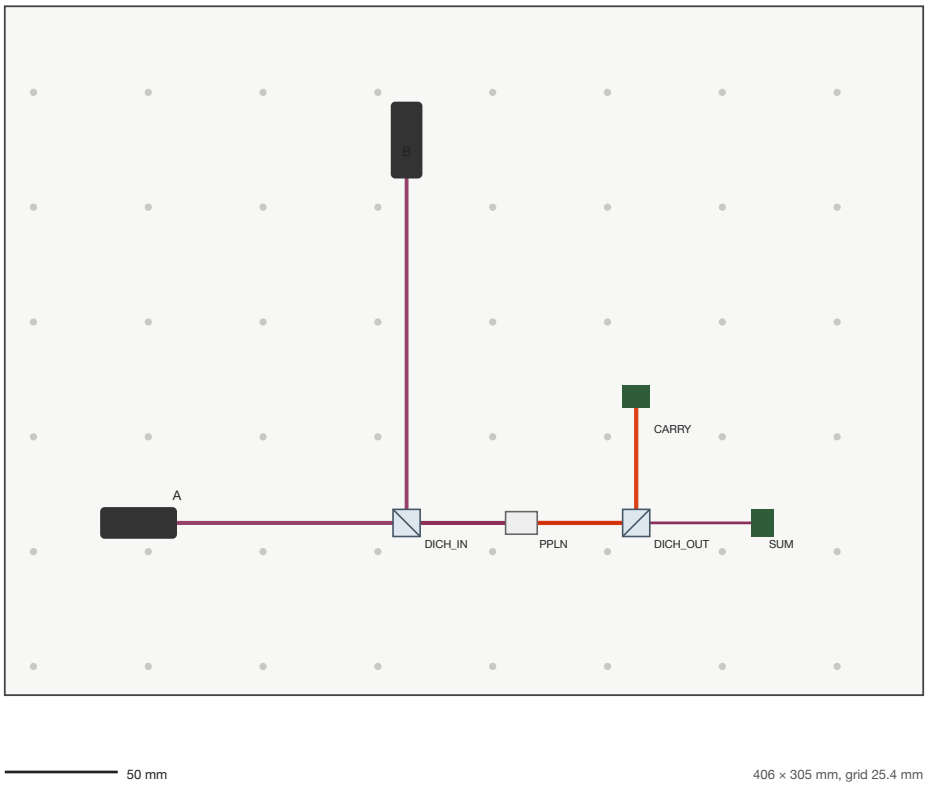


Figure 1: Plan view. Beams coloured by wavelength (infrared shown dark magenta), width ∝ log power; dashed = leaves the plate; grey dots = table holes.

Key specifications

Quantity	Value
Wavelengths	1064 nm, 1550 nm
Input power	368 mW from 2 source(s)
Delivered (fibers + detectors)	368 mW
Elements / beam segments	5 / 10
Baseplate	406.4 × 304.8 × 25.4 mm, 4 printed tile(s)

Verification

PASS score 100 / 100 (Beamline tracer: 2D ray + Gaussian beam + Jones polarization)

Targets

Target	Status
met: CARRY: 347.400 mW (need ≥ 100.000)	met

Checks

Severity	Check	Message
INFO	target	met: CARRY: 347.400 mW (need ≥ 100.000)

Bill of materials

Purchased parts

Part	Description	Vendor	Qty	Unit	Line
DMLP1180	Ø1" longpass dichroic, 1180 nm cutoff — 1550 reflect / 1064 pass	Thorlabs	1	$190.00	$190.00
DMLP805	Ø1" longpass dichroic, 805 nm cutoff — 631 nm reflect / IR pass	Thorlabs	1	$190.00	$190.00
KM100	kinematic mirror mount, Ø1" × 2: DICH_IN, DICH_OUT	Thorlabs	2	$48.00	$96.00
PDA20CS2	InGaAs amplified photodetector, 800-1700 nm — SUM	Thorlabs	1	$480.00	$480.00
PDA36A2	Si amplified switchable-gain photodetector — CARRY	Thorlabs	1	$360.00	$360.00
DPSS-1064-300	laser, 1064 nm, 218.5 mW — A		1	—	—
FPL1055T	laser, 1550 nm, 150 mW — B		1	—	—
MSFG1-0.5	PPLN crystal for SFG/SHG + oven [specialist; QPM period per process] — PPLN waveguide, QPM for 1064 +…	Covesion	1	$1,900.00	$1,900.00

Estimated total: $3,241.00

list-price estimates, USD, excluding lasers; verify with vendors

Hardware

Part	Description	Qty	Unit
SH25S050	1/4-20 x 1/2" cap screw (25 pack -> per piece)	48	$0.40
SH8S100	8-32 x 1" cap screw (per piece)	5	$0.30
DOWEL-2x8	Ø2 x 8 mm steel dowel pin (per piece)	10	$0.15
M3x25+NUT	M3 x 25 mm cap screw + nut (tile lap joints)	14	$0.20

3D-printed parts

Part	STL file	cm³	Print h	PLA g
baseplate_r1c1	stl/baseplate_r1c1.stl	894.9	24	357
baseplate_r2c1	stl/baseplate_r2c1.stl	577.1	15.6	233
baseplate_r1c2	stl/baseplate_r1c2.stl	949.1	25.4	378
baseplate_r2c2	stl/baseplate_r2c2.stl	621.4	16.8	250

Assembly

Step 1 **Print the baseplate** — Print 4 file(s) from stl/ flat, top side up: PLA, 20 % infill, 3 walls (≈ 81.8 h, ≈ 1218 g).

Step 2 **Join the tiles** — Lay the tiles out by their r/c names, overlap the 20 mm half-lap joints and fasten with 14 M3 × 25 bolts + nuts.

Step 3 **Press in the dowel pins** — Press 10 Ø2 × 8 mm pins into the paired pin holes; they key each mount's orientation.

Step 4 **Bolt to the table** — Fix the plate with 48 1/4-20 (or M6) cap screws through the counterbored holes (2-grid pitch, half a grid in from the edges).

Step 5 **DICH_IN** — Mount 1550 reflect / 1064 pass (DMLP1180, KM100) at site 1, (177.8, 76.2) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Outputs: transmitted → +x (218 mW); reflected → +x (150 mW). Reflects above 1300 nm.

Step 6 **PPLN** — Mount PPLN waveguide, QPM for 1064 +… (MSFG1-0.5) at site 2, (228.6, 76.2) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Outputs: transmitted → +x (12.5 mW); reflected → +x (8.59 mW); sum-frequency → +x (347 mW)

Step 7 **DICH_OUT** — Mount 631 nm reflect / IR pass (DMLP805, KM100) at site 3, (279.4, 76.2) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Outputs: transmitted → +x (12.5 mW); transmitted → +x (8.59 mW); reflected → +y (347 mW). Reflects below 800 nm.

Step 8 **SUM** — Mount SUM (PDA20CS2) at site 4, (330.2, 76.2) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Expect ≈ 21.1 mW on the detector.

Step 9 **CARRY** — Mount CARRY (PDA36A2) at site 5, (279.4, 127.0) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Expect ≈ 347 mW on the detector.

Alignment procedure

Align in beam order with an iris and an IR card; adjust one knob at a time and keep the best reading.

Order:

No tuner run attached: use the design values listed in the assembly steps.

Expected readings

- expect ≈ 347 mW at CARRY
- expect ≈ 21.1 mW at SUM

Laser safety

Wavelengths: 1064 nm, 1550 nm. **Maximum source power on the table:** 218 mW.

Wavelength	Band	Eyewear (generic guidance)
1064 nm	near-IR (invisible!)	OD ≥ 4 at 1064 nm (generic; confirm with your laser safety officer)
1550 nm	IR (eye-safe band, still hazardous to skin/optics)	OD ≥ 4 at 1550 nm (generic; confirm with your laser safety officer)

Beam dumps

No beam dumps in this design.

Verify all beam dumps before full power. Align at the lowest usable power, never look along the beam, remove watches and rings, keep the beam below eye level, and enclose invisible (IR) beams.

Credits

- Layout concept and “paint by numbers” baseplates: PyOpticL (code-to-CAD optics).
- Designed by Gemini `gemini-4-argon` agents, verified by the Beamline tracer.
- Prices are list-price estimates; part numbers should be checked against current vendor catalogs.